

Quranic Chatbot

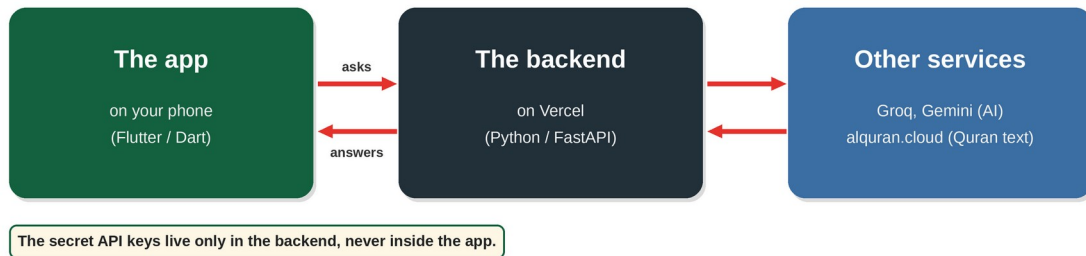
Code Guide (Simple Edition)

The main things about how the app works

Read time: about 10 minutes

1. What is in the project

The project has **two parts** that talk to each other.



Part	Where	What it does
The app	<code>flutter_app/lib/</code>	Everything you see and tap on the phone
The backend	<code>backend/api/index.py</code>	Keeps the secret keys, talks to the AI, matches recitations to verses

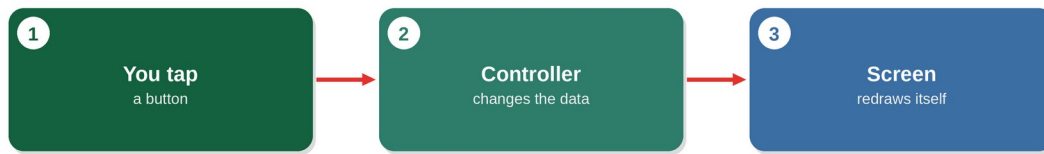
2. The app's folders

All app code is inside `lib/`. Each folder has one clear job.

Folder	What is inside
<code>screens/</code>	The pages you see (Home, Surah reader, Chat, Settings...)
<code>widgets/</code>	Small reusable pieces (verse card, buttons, audio bar)
<code>state/</code>	The "controllers" that remember data (settings, chat, bookmarks)
<code>services/</code>	The code that talks to the outside: backend, audio, saved files
<code>models/</code>	Simple data shapes: Surah, Verse (Ayah), Chat message
<code>l10n/</code>	All text in English, Urdu and Arabic
<code>theme/</code>	Colours and the 12 themes
<code>main.dart</code>	The starting point. It also lists every other file.
<code>services.dart</code>	The backend address (you must edit this)

3. How a tap becomes a change on screen

Every feature works the same way.



Screens only show things. Controllers (in the state/ folder) hold the data.

For example, when you change the language in Settings, the controller saves it and the whole app redraws in the new language.

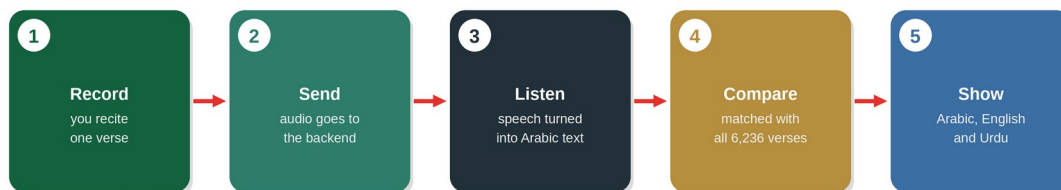
4. The main features

AI chat



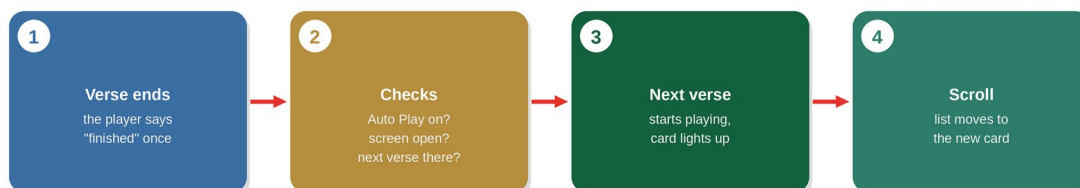
The answer comes with verse references. The app looks up each reference and shows the **real Quran text**, not text written by the AI.

Recite & Recognize



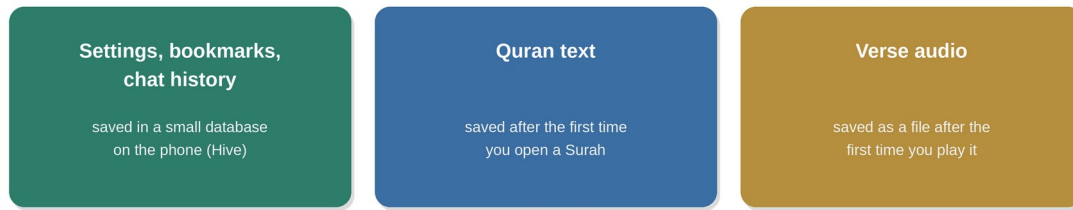
This is **not an AI guess**. The backend compares what it heard with the real Quran text. If the match is weak, it says “no match”.

Auto Play



Auto Play is **off by default**. Turn it on in Settings.

Offline use



Nothing here is sent to a server. Uninstalling the app deletes it.

Because of this, verses you have already opened or played still work without internet.

5. The backend in short

The backend is one file. It has these jobs:

Address	Job
<code>/surahs, /surahs/{id}/ayahs, /juz/{id}</code>	Give Quran text with English and Urdu
<code>/search</code>	Search verses
<code>/chat</code>	AI answer (Groq first, Gemini if Groq fails)
<code>/stt</code>	Turn your voice into text
<code>/recognize</code>	Find which verse was recited

It needs two secret keys, `GROQ_API_KEY` and `GEMINI_API_KEY`, set in Vercel.

6. Where do I change...?

I want to change	Open this file
The backend address	<code>lib/services.dart</code>
Any text or language	<code>lib/l10n/strings.dart</code>
Colours or themes	<code>lib/theme/theme_catalog.dart</code>
How a verse card looks	<code>lib/widgets/verse_widgets.dart</code>
Auto Play	<code>lib/screens/surah_screen.dart</code>
Default settings	<code>lib/state/settings_controller.dart</code>
The AI models or the chat prompt	<code>backend/api/index.py</code>

7. Words used in this guide

Word	Meaning
Widget	A piece of the screen, like a button or a card

Word	Meaning
Controller	Holds data and tells the screen when it changes
Service	Code that does outside work: network, audio, saving
Backend	The small server on Vercel
Hive	The small database inside the app
Ayah	One verse of the Quran